# Toronto Metropolitan University

Department of Electrical, Computer, and Biomedical Engineering

Midterm Examination: Electric Circuit Analysis (ELE202)

Duration: 2 Hours

July 2025

Student's Name: [redacted]

[First Name, Last Name]

Student Number: [redacted]

## EXAM INSTRUCTIONS

**Resources Allowed:**

This is a closed-book and closed-lecture-notes exam. You may only use a single-sided handwritten A4 formula sheet and an approved calculator.

**Structure:**

The exam consists of 5 questions, each with multiple parts. Answer all the questions. Marks for each part are indicated, and final numerical answers carry specific weight.

**Working & Organization:**

- Show all steps clearly.
- If you need extra space, use the back of each page and clearly label which question/part you're answering.
- No additional paper will be provided.

**Clarifications:**

- No questions will be answered during the exam unless it's urgent.
- If a question seems unclear, include a brief statement of any assumptions you made in your answer.

Table: Score Sheet

| Question | Marks | Obtained Marks |
|---|---|---|
| Q1 A | 10 | 10 |
| Q1 B | 10 | 7 |
| Q2 | 20 | 16 |
| Q3 | 20 | 17 |
| Q4 | 20 | 10 |
| Q5 | 20 | 6 |
| Total | 100 | 66 |

Q1 (A) **Figure 1** shows a graph of current versus time (not to scale). Sketch the corresponding charge versus time curve on the same axes using this graph.

$q = \int i\,dt$ = area

$0-2 = \frac{1}{2}(2)(2) = 2$

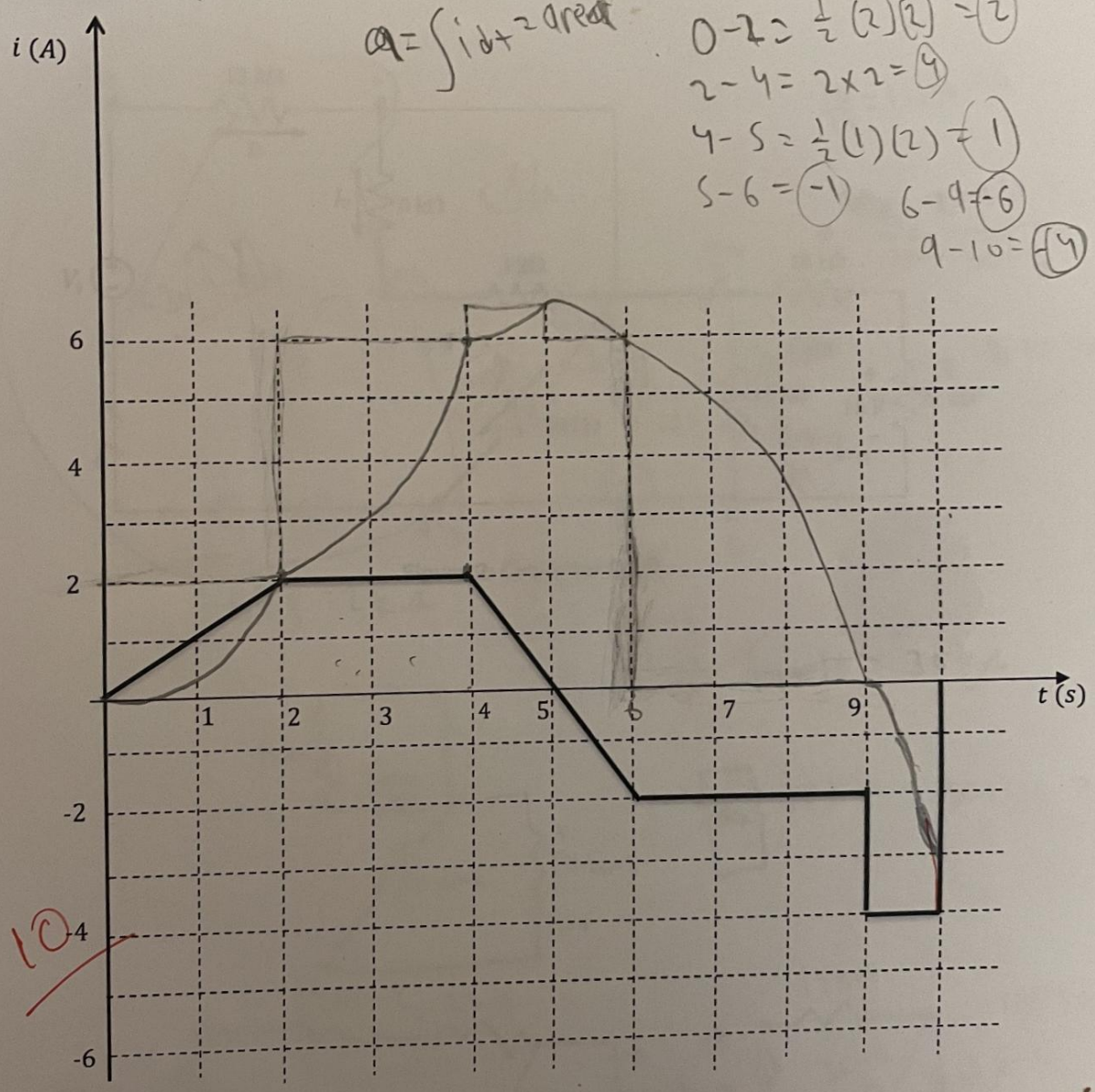
$2-4 = 2\times 2 = 4$

$4-5 = \frac{1}{2}(1)(2) = 1$

$5-6 = -1$

$6-9 = -6$

$9-10 = -4$




10

**Figure 1**: Figure for Q1(A)

$-V_s + 12i_1 +$

Q1 (B) Using the **circuit reduction method** (i.e., calculating equivalent resistance), determine the values of $I_s, I_0, I_x$, and $V_s$ for the circuit shown in **Figure 2**. Also, calculate the equivalent resistance of the circuit.

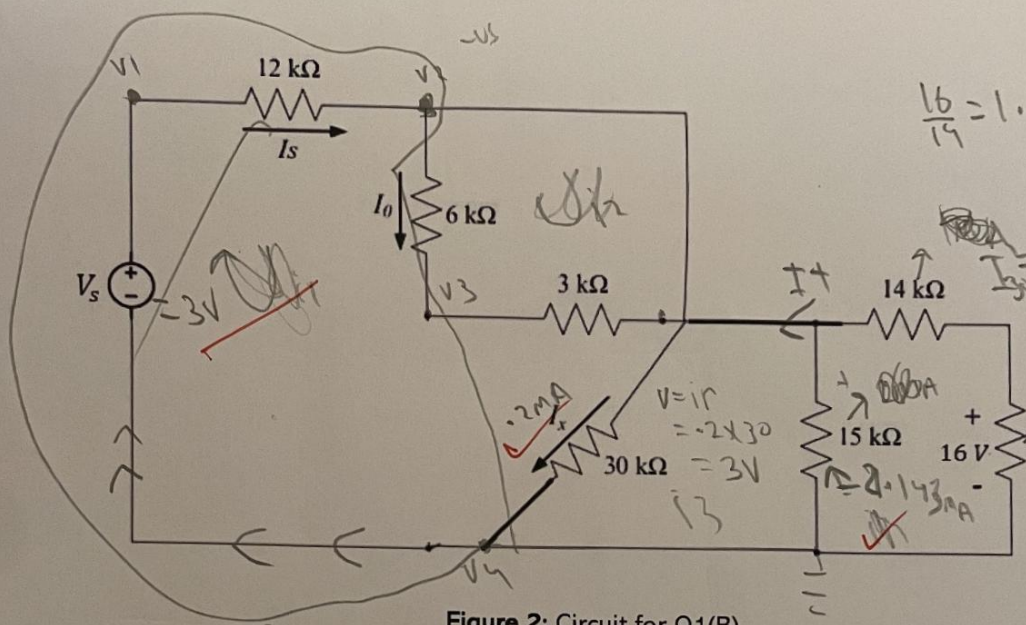
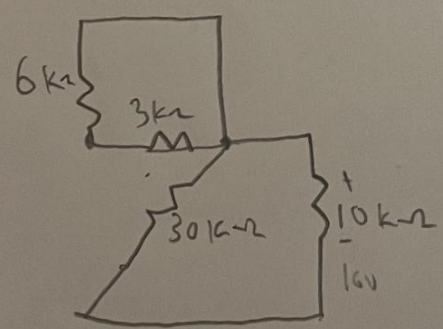
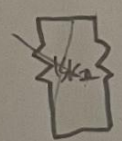



Figure 2: Circuit for Q1(B)

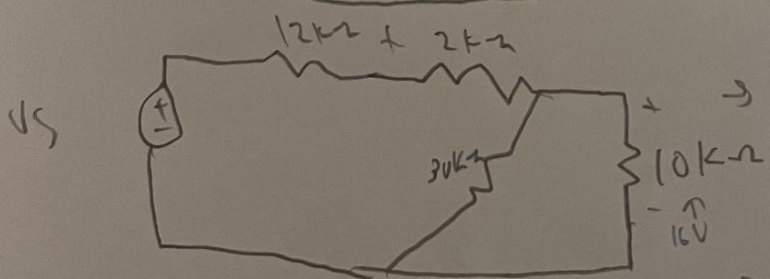
6||3 = 2kΩ

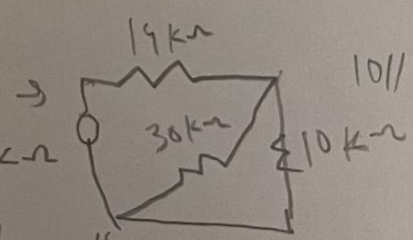
14+16 = 30kΩ



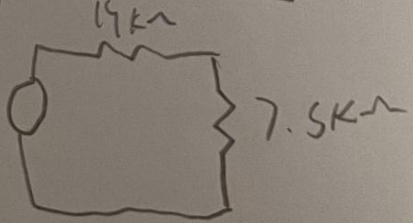




30||15 = 10kΩ

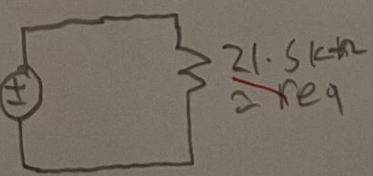



10||30 = 7.5kΩ




14+7.5 = 21.5kΩ




$I_0 = \frac{V_s}{21.5k\Omega}$

$\frac{V2}{8} - 2VX - \frac{V1-V2}{2} + \frac{V3}{10} = 12 \rightarrow -\frac{1}{2}V1 + 0.625V2 - 2VX = 1.2$

$V3 - V2 = 12V \rightarrow 12 - V2 = 12 \rightarrow 12 - 12 = V2 = 0$

$V1)\ \frac{V1}{6} - \frac{24-V1}{3} + \frac{V1-V2}{2} + 2VX = 8$

$\frac{1}{6} + \frac{1}{3} + \frac{1}{2} = .999$

**Q2.** Solve the circuit in **Figure 3** using the **Nodal Analysis** method (including Supernode if necessary). Assume the ground node is at the bottom of the circuit.

(A) Write the nodal equations for all non-reference (non-ground) nodes. (8 marks)

$.999V1$

**(B)** Simplify the equations from part (A). (4 marks)

**(C)** Solve the simplified system from part (B) and provide the **numerical values** of the node voltages. (4 marks)

**(D)** Express the **voltage across the 2 Ω resistor** and the **current through** 10 Ω **resistor** in terms of node voltages. Then, calculate and provide their **numerical values**. (4 marks)

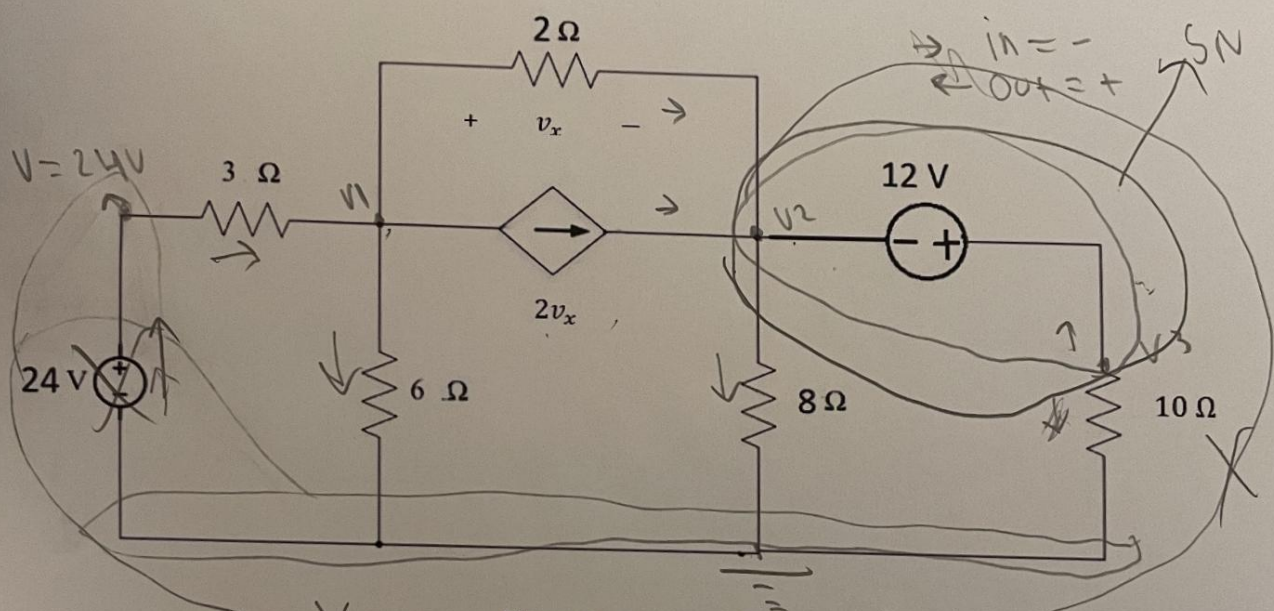



**Figure 3:** Circuit for Q2

$SN)\ \frac{V3}{10} + \frac{V2}{8} - \frac{V1-V2}{2} - 2VX = 0$

inside SN): $V3 - V2 = 12V$

$-V2 + V3 = 12V$

$V2 = 0V \quad V3 = 12V$

$V1)\ \frac{V1-V2}{2} + \frac{V1}{6} - \frac{V1-24}{3} + 2VX = 0$

$.625V2$

$1/8 + 1/2$

16

① $-V1\left(\frac{1}{2}\right) + V2\left(\frac{1}{2} + \frac{1}{8}\right) + V3\left(\frac{1}{10}\right) = 0$

① $-\frac{1}{2}V1 + 0.625(V2) + \frac{1}{10}V3$

$2VX \quad -\frac{1}{2}V1 + \frac{1}{10}V3 = 7.5$

② $-V2 + V3 = 12V$

$-12 + V3 = 12$

③ $V1\left(\frac{1}{2} + \frac{1}{6}\right) - 2V2 + \frac{V1}{3} = 8$

$.333V1 - \frac{1}{2}V2 + 2VX = 8$

$.333V1 + 2VX = 4$

Q3. For the circuit shown in **Figure 4**, use **Mesh Analysis** (assume all mesh currents flow in the **clockwise direction**) to solve the following:

**(A)** Write the mesh (loop) equations for all independent clockwise mesh currents. (8 marks)

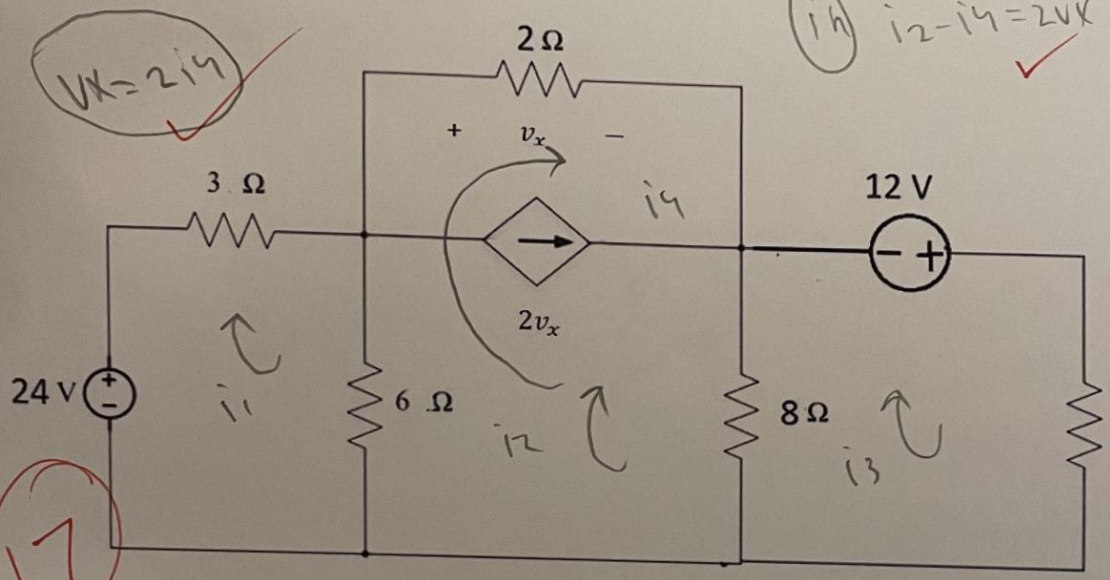
**(B)** Simplify the equations from part (A). (4 marks)

**(C)** Solve the simplified system from part (B) and provide the **numerical values** of the clockwise mesh currents. (4 marks)

**(D)** Express the **voltage across the 2 Ω resistor** and the **current through 10 Ω resistor** in terms of the clockwise mesh currents. Then, calculate and provide their **numerical values**. (4 marks)

$V_x = 2i_4$

$i_4$ $i_2 - i_4 = 2V_x$




$V = iR$
$= 12(10)$
$= 120V$

17

**Figure 4:** Circuit for Q3

(i1) $-24 + 3i_1 + 6(i_1 - i_2) = 0$

(i2+i3) $6(i_2 - i_1) + 2i_4 + 8(i_2 - i_3) = 0$

$\cancel{6i_2} - 6i_1 + 2i_4 + \cancel{8i_2} - \cancel{8i_3} = 0$

$= -6i_1 + 14i_2 - 8i_3 + 2i_4 = 0$

(i3) $8(i_3 - i_2) + 10i_3 = 12$

$8i_3 - 8i_2 + 10i_3 = 12$

(1) $9i_1 - 6i_2 = 24$ (2) $-6i_1 + 14i_2 - 8i_3 + 2i_4 = 0$ (3) $-8i_2 + 18i_3 = 12$

(4) $i_2 - i_4 = 2V_x$

$i_2 = 2i_4 + i_4$

$i_2 = 3i_4$

$i_2 - 3i_4 = 0$

$+36i_4 -$

$i_1 = 9.67A$ $i_2 = 10.5A$ $i_3 = 12A$ $i_4 = 3.5A$

$V_x = (3.5)(2)$

$7A$

$2V_x = 14A$

**Q4.** Use the **superposition principle** to determine the voltage across the terminals $V_{ab}$ for the circuit in **Figure 5**.

a. Superposition Configurations (2 marks)

- Draw the two separate circuit configurations required for superposition analysis.

b. Voltage Due to 6A Current Source (8 marks)

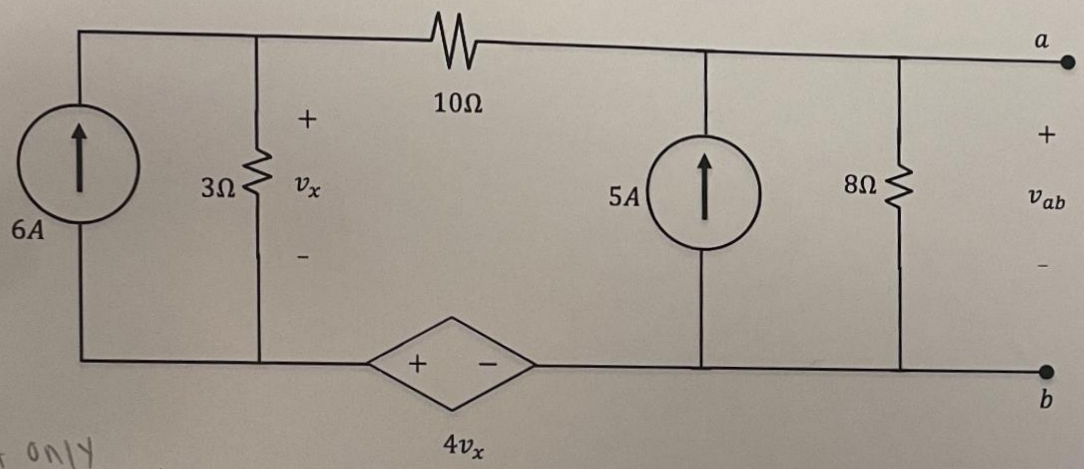
- Using any appropriate circuit analysis method, calculate the voltage $V_{ab}$ due to the 6A current source only.

c. Voltage Due to 5A Current Source (8 marks)

- Using any suitable method, determine the voltage $V_{ab}$ due to the 5A current source only.

d. Total Voltage $V_{ab}$ (2 marks)

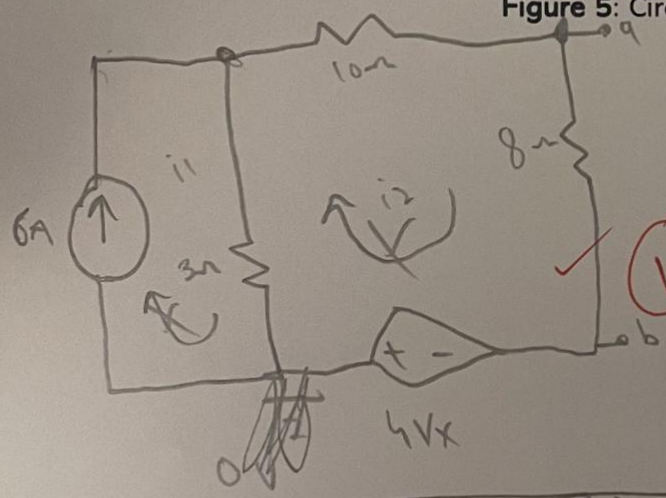
- Calculate the total voltage $V_{ab}$ for both current sources.




$V_x = 3 \times 6 = 18V$

6A only

**Figure 5**: Circuit for Q4




$i_1 = 6A$

$i2)\ 3(i_2 - i_1) + 10i_2 + 8i_2 - 4V_x = 0$

$3(i_2 - 6) + 10i_2 + 8i_2 - 4(18) = 0$ ?

$3i_2 - 18 + 10i_2 + 8i_2 = 72$

$21i_2 = \frac{90}{21}$ 4

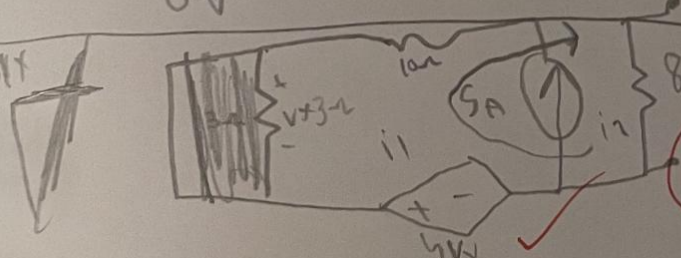
$i_2 = 4.3A$

$V_{ab} = (4.3)(8) = 34.4V$

1

$34.4 - 7.44 = 30V$

5A only




$8\Omega\ (i_1 + i_2) - 3i_1 + 10i_1 + 8i_2 - 4V_x = 0$

$3i_1 + 8i_2 - 4(3i_1) = 0$

$-12i_1 = 0$

(inside) $= i_2 - i_1 = 5$ 4

$V_x = 3i_1$

$i_1 = -4.44A$

$i_2 = .555A$

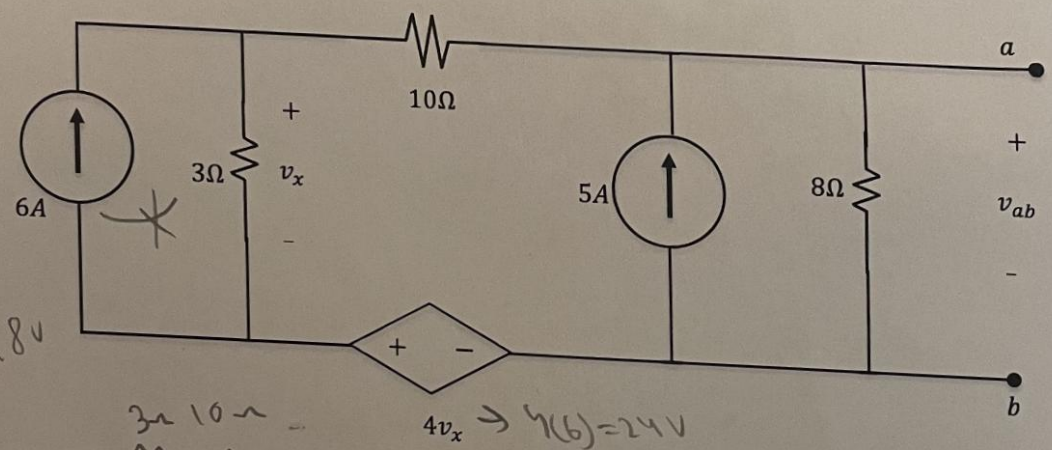
1 $i_1 + 8i_2 = 0$

$8 \times (.55) = 4.44V$

1

Q5. For the circuit in **Figure 6**, use **Thevenin's Theorem** to simplify the circuit.

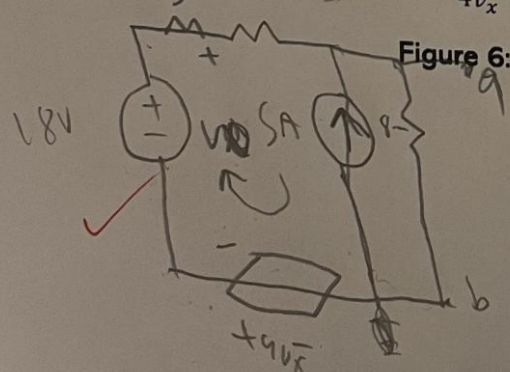
1. **Thevenin Equivalent Circuit (14 marks)**
   - Determine the **Thevenin equivalent voltage $V_{th}$** across terminals $a$ and $b$ using **the source transformation technique.**
   - Determine the **Thevenin equivalent resistance $R_{th}$** seen from the terminals $a$ and $b$.
2. **Maximum Power Transfer Condition (2 marks)**
   - State the condition for **maximum power transfer** to the load resistor $R_L$ connected between terminals $a$ and $b$. Calculate the value of $R_L$ that satisfies this condition.
3. **Maximum Power Delivered to the Load (2 marks)**
   - Using the Thevenin equivalent circuit, calculate the **maximum power** that can be delivered to the load resistor $R_L$.
4. **Circuit Diagram (2 marks)**
   - Draw the **Thevenin equivalent circuit** with the load resistor $R_L$ connected.




6×3=18V

4(6)=24V

**Figure 6:** Circuit for Q5




$V_x = 3i_1 = -2.444 \times 3 = -7.32V$

$V_a = -18 + 3i_1 = 0 \rightarrow 3i_1 = \frac{18}{3} = 6V$

1) $-18 + 3i_1 + 10i_1 + 8i_2 - 4V_x = 0$

~~$13i_1 + 8i_2 = 18$~~

$13i_1 + 8i_2 - 4(3i_1) = 18$

$13i_1 + 8i_2 - 12i_1$

① $i_1 + 8i_2 = 18$

② $-i_1 + i_2 = 5$

$i_1 = -2.444A$

$i_2 = 2.555A$

4